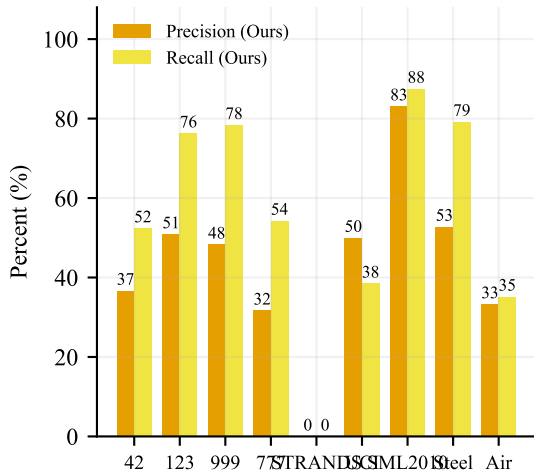
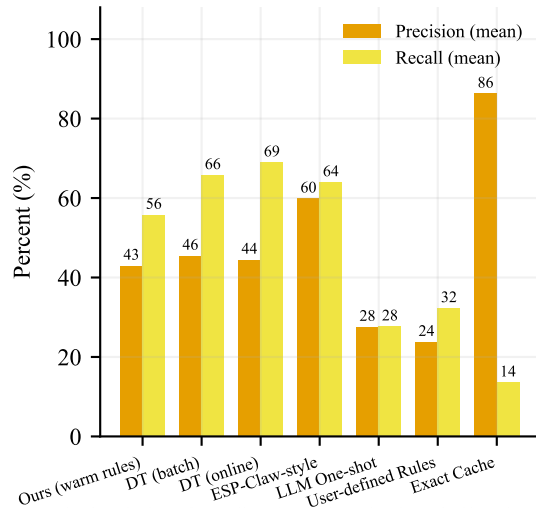


(a) Ours — teacher-relative P/R (warm rules)



(b) Method comparison (mean over 9 datasets)



Provenance: 60 sensor snapshots per dataset; ground truth = majority of 3 teacher (DeepSeek T=0) repeats; precision = $P(\text{method action} = \text{teacher} \mid \text{method acted})$; recall = $P(\text{method acted and matched} \mid \text{teacher acted})$. Replaces the Phase-0 placeholder (no user feedback).